

3D Plant Segmentation Protocol using CloudCompare

1. Prerequisites & Setup

1.1. Install CloudCompare: <https://cloudcompare.org/index.html>

1.1.1. Locate the source folder containing the plant point clouds (.pcd)

1.1.2. Create the following folder structure:

1.1.2.1. **\Raw Data\Processed Data** -> Main process folder.

1.1.2.2. **\Raw Data\Processed Data\Original** -> Raw, untouched file.

1.1.2.3. **\Raw Data\Processed Data\Label** -> File exported from Cloud Compare that contains the classification (4 labels).

1.1.2.4.

Label Index	Object
0	Stem
1	Leaf
2	Stake
3	Background

2. Segmentation Process

2.1. Step A: File Preparation

2.1.1. Open CloudCompare.

2.1.2. Open a point cloud file from the source folder.

2.1.3. Click on the original point cloud, go to the main toolbar, click on the clone tool (two sheep icon). This is done to make sure we have a clean backup.

2.2. Step B: The Segmentation Task

2.2.1. Select the plant point cloud in the DB Tree window on the left.

2.2.2. Click the "Segment" tool in the main toolbar (it looks like a pair of scissors).

2.2.3. You will now be in segmentation mode. Use your mouse to draw a polygon around the first plant part you want to separate (e.g., a single leaf, or part of the pot). Start by removing the part of the plant that is easier to segment. You may want to start by removing the pot and the background as this is the easiest of all the parts in the plant; however, every plant will be different, so there's not a real sequence to follow.

2.2.3.1. Left click to add points to your polygon.

2.2.3.2. Right click to close the polygon.

2.2.3.3. You can segment the parts of the plant either by using "Segment In" or "Segment Out". Just make sure to save the result and label it accordingly.

2.2.3.3.1. **Segment in:** Once the polygon is closed, a new toolbar will appear. Click the "Segment In" button (the icon with the scissors inside a polygon). This will create a new point cloud containing only the points you selected.

2.2.3.3.2. **Segment out:** Click the "Segment Out" button (scissors outside a polygon) to create a cloud of everything you didn't select. You will now have two new clouds in your DB Tree.

2.2.4. Exit the segmentation tool by clicking the checkmark or pressing the ESC key.

2.2.5. Rename the new cloud. Find the new cloud in the DB Tree (it will be named something like *[OriginalName].segmented*). Click on it, press F2, and rename it according to this name convention:

2.2.5.1. For leaf: "id.leaf_n"; where n is the leaf number. (e.g., "2024-05-27-20-24.leaf_1"

2.2.5.2. For stem: "id.stem"

2.2.5.3. For stake: "id.stake"

2.2.5.4. For background: "id.background_n"; where n is the background number. (e.g., "2024-05-27-20-24.background_1"

2.2.5.5. Now, at this point you will have multiple "id.leaf_n" and "id.background_n" pointclouds. You want to merge them into one single "leaf pointcloud" and "background pointcloud". To do this, you need to select all of them, and then click on the "merge" button up top, located in the toolbar (the one with the blue figure and the arrow pointing to the left). You'll end up with one pointcloud. Rename this "id.leafs" and "id.background". This is your leaf and background classified pointcloud.

2.2.6. Now that you have successfully segmentized all parts of the plants, and merged the leafs and background, it's time to assign a label to the created segment:

2.2.6.1. Go to Edit -> Scalar fields -> Add classification SF -> class number (e.g., leaf = 1) -> OK. Repeat the same process to add a label to the background, stem, and stake (check label index above).

2.2.7. Finally, we are going to merge all segmentized parts of the plants into one single point cloud named "id.plant_classified".

2.2.8. Then we will proceed to use the same "id.plant_classified" file and make a copy of this by cloning it. Select the file, and then click on the button in the upper toolbox that has 2 sheep in it. Then, click on the cloned pointcloud, and go to Edit -> Scalar fields -> Delete all (!). This will remove all labels from this pointcloud.

2.2.8.1. Rename the file to "id.plant_unclassified". This is your new raw .pcd file.

2.3. Step C: Finalizing and Saving

2.3.1. Click on the "id.plant_classified", go to File -> Save -> Save as type (Use the drop-down menu) and saved as "ASCII Cloud (.txt)". Save it in the folder "\\Raw Data\\Processed Data\\Label"

2.3.2. Click on File -> Save project", to create a final save of your .bin project file.

3. Annotation Rules & Guidelines

3.1. What to Segment: All distinct plant parts must be segmented into their own clouds. This includes:

3.1.1. Each individual leaf.

3.1.2. The main stem.

3.1.3. The stake.

3.1.4. The background

4. Naming Convention: This is critical for consistency. Please use the following format: *partName_##*.

4.1. Examples: leaf_1, leaf_2, leaf_3, background_1.

5. Boundary Rules:

5.1. The points where two parts meet should be carefully separated. For example, the stem segment should end where the petiole segment begins. Try to make the split at the natural joint.

6. **Completeness:** Every point from the original plant cloud must be assigned to a part. If you have unassigned points left over, determine if they are part of the plant or background and assign them the respective classification to it.